

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

Enter your answer

8

What is the correct order of operations of the OpenGL graphics pipeline?
(1 Point)

modelview transformation

viewport transform

projection transformation

divide by w

1

4

2

3

9

$$\begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & 1.5 & 0 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Write the OpenGL code segment that generates the following transformation matrix.

(1 Point)

Enter your answer

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glScalef(-2.0, 1.5, -3.0);
```

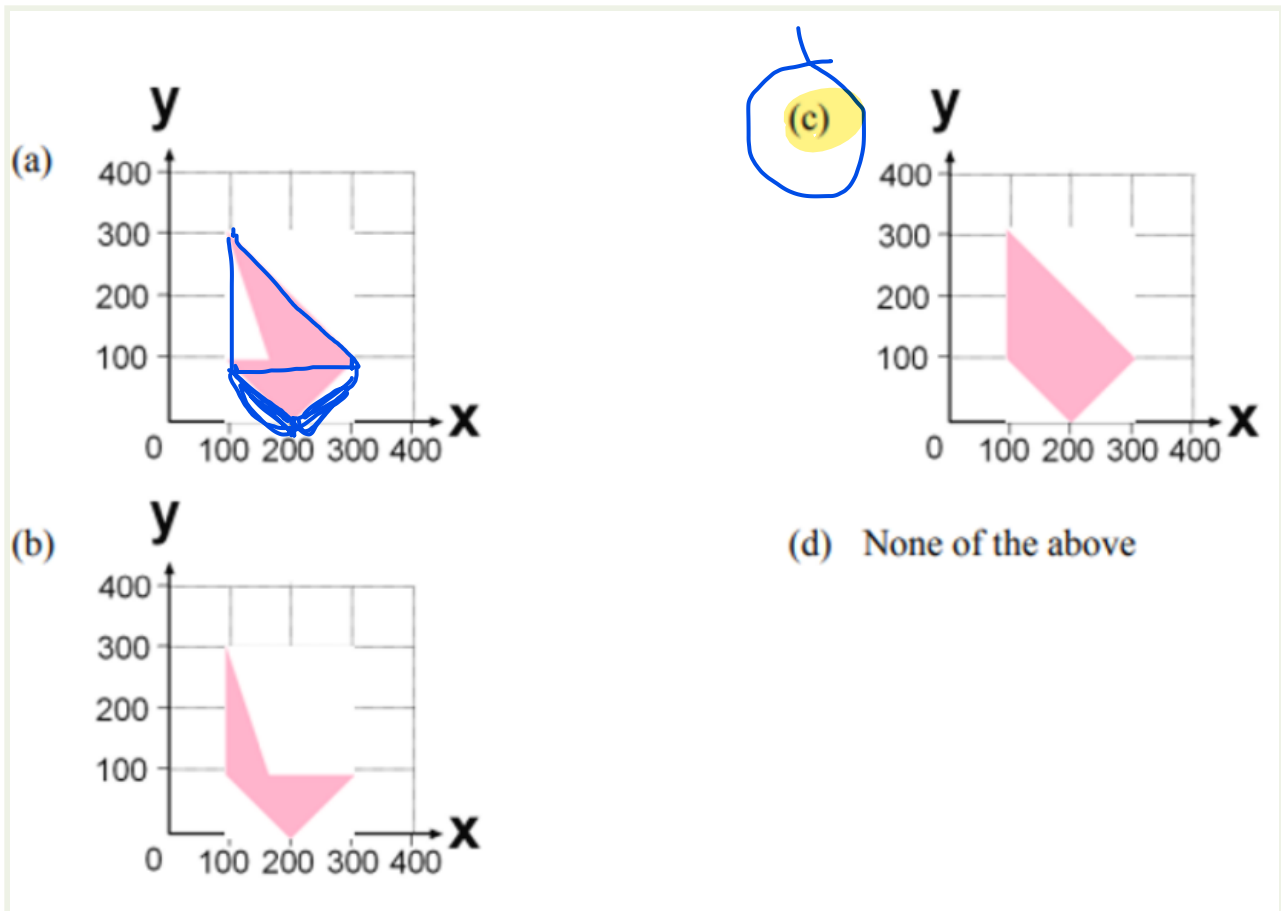
10

Given is an OpenGL program with the display method

```
void display(void) {
glClear(GL_COLOR_BUFFER_BIT);
glBegin(GL_TRIANGLE_FAN);
glVertex2f(300, 100);
glVertex2f(100, 300);
glVertex2f(100, 100);
glVertex2f(200, 0);
glEnd();
glFlush(); }
```

What shape does this display method draw?

(1 Point)



☐ (a)

☐ (b)

☐ (c)☐ (d)

11

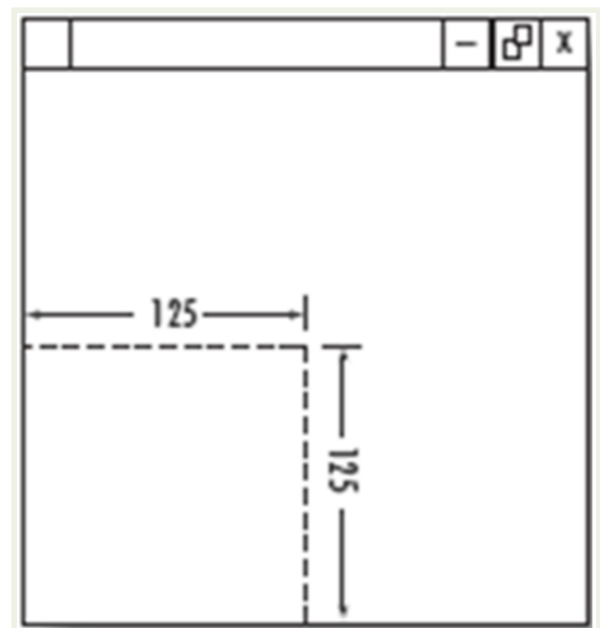
Culling

..... is the term used to describe the technique of eliminating geometry that we know will never be seen.

(1 Point)

Enter your answer

12



`glViewport(0,0,125,125)`

Write the OpenGL code segment that sets the output to the following square.

(1 Point)

Enter your answer

13

For a rotation matrix R and a translation matrix T , $RT = TR$, even though matrix multiplication is not generally commutative.

(1 Point)

☐ True

☒ False

14

Find the equation of the plane defined by the three points $A=[1 \ 2 \ -1]$, $B=[4 \ 2 \ 3]$, and $C=[1 \ 4 \ -1]$

(1 Point)

Enter your answer

$$-8x + 6z = -14$$

15

Rotate (z-axis, 90) * Rotate (z-axis, 45) = Rotate (z-axis, 45) * Rotate (z-axis, x1), x1 =

(1 Point)

Enter your answer

90

16

The frame buffer is a region of memory sufficiently large to hold all pixels of the display. The OpenGL command, clears a region in it which corresponds to the current display window.

(1 Point)

`glClear(GL_COLOR_BUFFER_BIT)`

Enter your answer

17

What is the normalized surface normal vector to the plane $x+2y+2z+4=0$?

(1 Point)

Enter your answer

 $(1/3, 2/3, 2/3)$

18

Rotating the 3D point (1,0,0) by -90 around z-axis gives what:

(1 Point)

☐ (0,-2,0)☐ (-1,0,0)☐ (0,1,0)☒ (0,-1,0)

19

Convert the following Target Camera to Free Camera

Eye = (2, 0, 0)

Center = (1, 0, 0)

Up = (0, 1, 0)

$F=C-E=(-1,0,0)$

(1 Point)

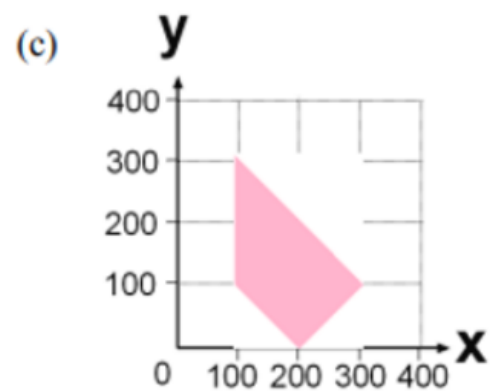
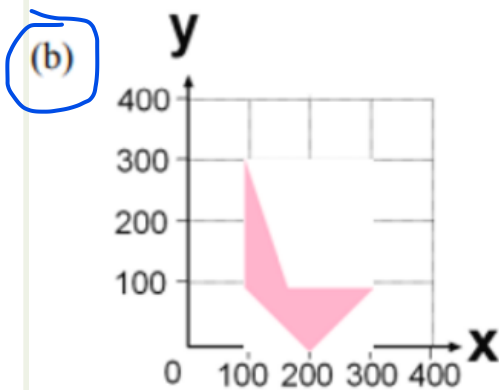
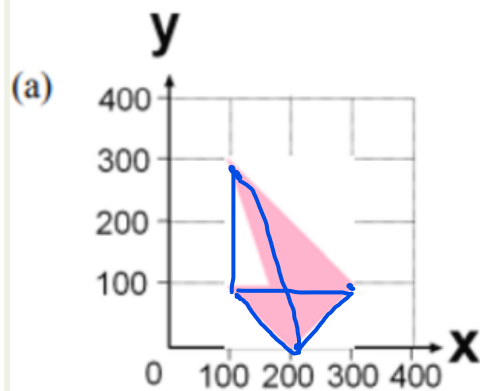
Enter your answer

20

Given is an OpenGL program with the display method

```
void display(void) {  
glClear(GL_COLOR_BUFFER_BIT);  
glBegin(GL_TRIANGLE_STRIP);  
glVertex2f(300, 100);  
glVertex2f(200, 0);  
glVertex2f(100, 100);  
glVertex2f(100, 300);  
glEnd();  
glFlush(); }
```

What shape does this display method draw?
(1 Point)



(d) None of the above

☐ (a)

☒ (b)

☐ (c)

☐ (d)

21

The RGB of a yellow color is
(1 Point)

☒ (1.0,1.0,0.0)

☐ (1.0,0.0,1.0)

☐ (0.0,1.0,1.0)

☐ (1.0,1.0,1.0)

22

The name of the 3D graphics API used in this course is
(1 Point)

[OpenGL](#)

Enter your answer

23

The 3x3 matrix for a translation by 2 in x and 3 in y is what?
(1 Point)

A. $\begin{pmatrix} 2 & 0 & 1 \\ 0 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ C. $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$ D. $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$

☐ A

☐ B

☐ C

☒ D

24

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Write the following transformation matrix as a one dimensional 16-entry array used in OpenGL
(1 Point)

Enter your answer

1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 3, 0, 0, 0, 0, 1

25

We use an additional toolkit named, which contains high-level functions for window creation, event handling and the drawing of solid objects.

(1 Point)

GLUT

Enter your answer

26

A Vector is a displacement/difference between two points

(1 Point)

- ☒ True
- ☐ False

27

How many triangles are drawn if we use the drawing mode `GL_TRIANGLE_STRIP` with six vertices between `glBegin(GL_TRIANGLE_STRIP)` and `glEnd()`?

(1 Point)

- ☐ 1
- ☐ 2
- ☐ 3
- ☒ 4
- ☐ Others